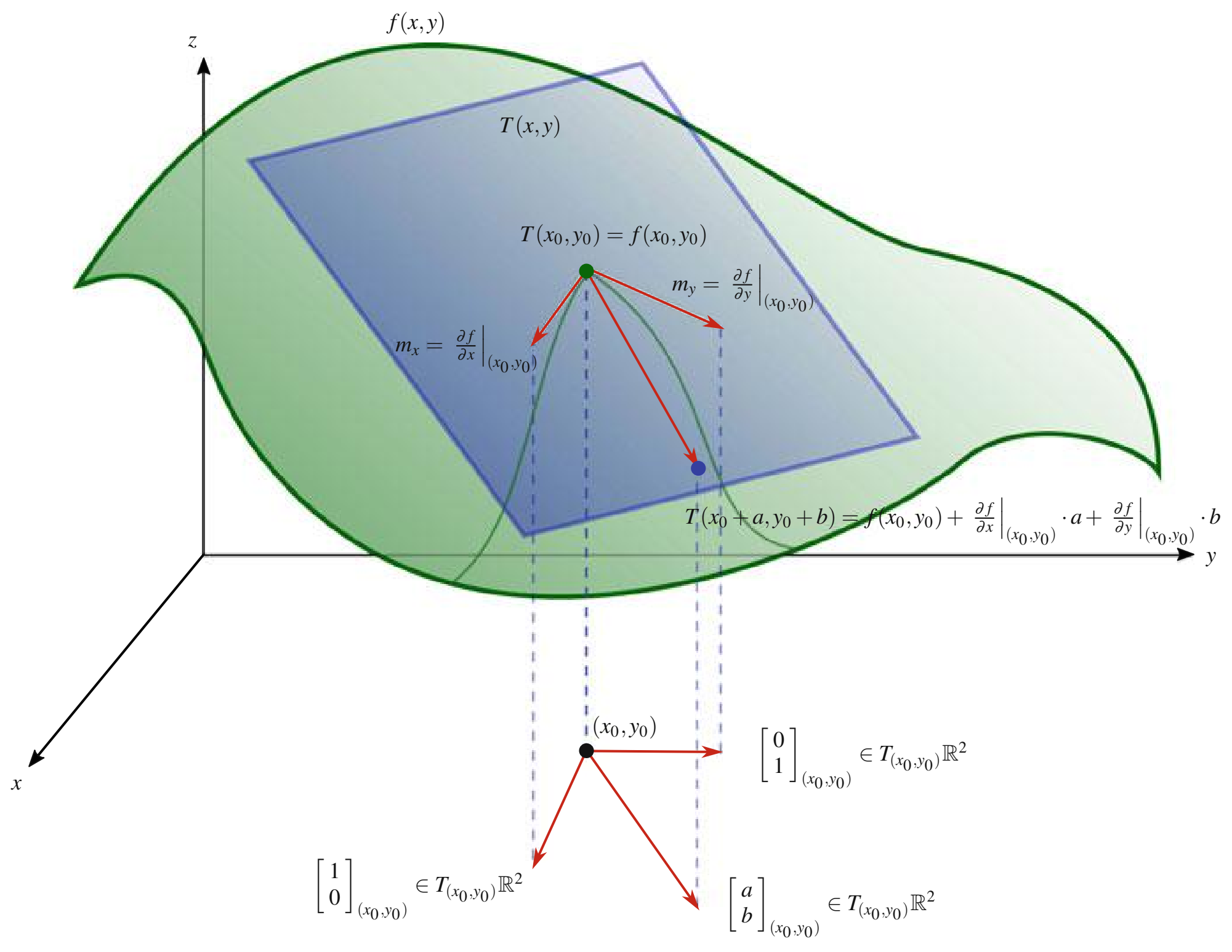

$f(x,y)$
$z$
$T(x,y)$
$T(x_0,y_0) = f(x_0,y_0)$
$m_y = \left.\frac{\partial f}{\partial y}\right|_{(x_0,y_0)}$
$m_x = \left.\frac{\partial f}{\partial x}\right|_{(x_0,y_0)}$
$T(x_0+a,y_0+b) = f(x_0,y_0) + \left.\frac{\partial f}{\partial x}\right|_{(x_0,y_0)} \cdot a + \left.\frac{\partial f}{\partial y}\right|_{(x_0,y_0)} \cdot b$
$y$
$x$
$(x_0,y_0)$
$\begin{bmatrix} 0 \\ 1 \end{bmatrix}_{(x_0,y_0)} \in T_{(x_0,y_0)}\mathbb{R}^2$
$\begin{bmatrix} 1 \\ 0 \end{bmatrix}_{(x_0,y_0)} \in T_{(x_0,y_0)}\mathbb{R}^2$
$\begin{bmatrix} a \\ b \end{bmatrix}_{(x_0,y_0)} \in T_{(x_0,y_0)}\mathbb{R}^2$